

STAT 5710 Final Project Abstract
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Project title: From Campus to Career: Measuring the Skills Gap Between T20 Graduates and Top Employer Demand across the 2024–2026 Labor Market

Keywords: skills gap analysis, labor market dynamics, predictive modeling, employer demand, graduate supply, O*NET, clustering, Revelio Labs, higher education

This project examines the question: How has the skills gap between what top employers demand and what tier 20 college (T20) grads offer changed from 2024 to 2026? Through an analysis of the temporal recent evolution of skills, we aim to determine mismatches between employer demand and graduate supply. To do so, we analyze the divergence between entry-level job postings from top employers and skills profiles of T20 (ranking classification by US News National Rankings 2026) university graduates across the two-year period of 2024 to the most recent landscape in 2026. Using Revelio Labs workforce data combined with O*NET occupational databases, we construct annual demand profiles from job postings and supply profiles from graduate LinkedIn profiles, computing skill-level gaps to identify which competencies show widening mismatches (which are emerging unmet needs) versus narrowing ones (successful adaptation to current demands). We use dimensionality reduction with principal component analysis (PCA), unsupervised clustering, and natural language processing (NLP) techniques to categorize skill groups, enriched with O*NET importance ratings to provide occupational context. This analysis will quantify where students from highly-selective educational institutions are successfully meeting top market needs and where critical talent shortages are emerging. Building on this analysis, the project culminates in a predictive model forecasting near future skill gap trajectories, which may be able to be used as a basis to offer actionable insights for curriculum design, career counseling, and workforce planning for T20 grads, and potentially other students entering the workforce in the near future across the United States.